

Lab Report-4

Md Ashraf

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Introduction:

In this lab, we explore several seismic migration methods, including **Stolt Migration**, **Gazdag (Phase-Shift) Migration**, and **Kirchhoff Migration**. Stolt Migration, based on Fourier transform principles, is particularly efficient for horizontally layered models and involves frequency-wavenumber ($f-k$) domain processing. The Gazdag (Phase-Shift) Migration, on the other hand, applies a phase shift to accommodate variable velocities, making it suitable for more complex geological settings. Kirchhoff Migration leverages ray-tracing to generate travel times, allowing it to handle intricate subsurface structures effectively.

Through the application of these methods to the `simple.su` dataset, this lab aims to illustrate how each migration technique enhances subsurface imaging. By comparing the results of different migration approaches, we gain insight into how each method influences the clarity and accuracy of the resulting seismic images.

1 Bash Script

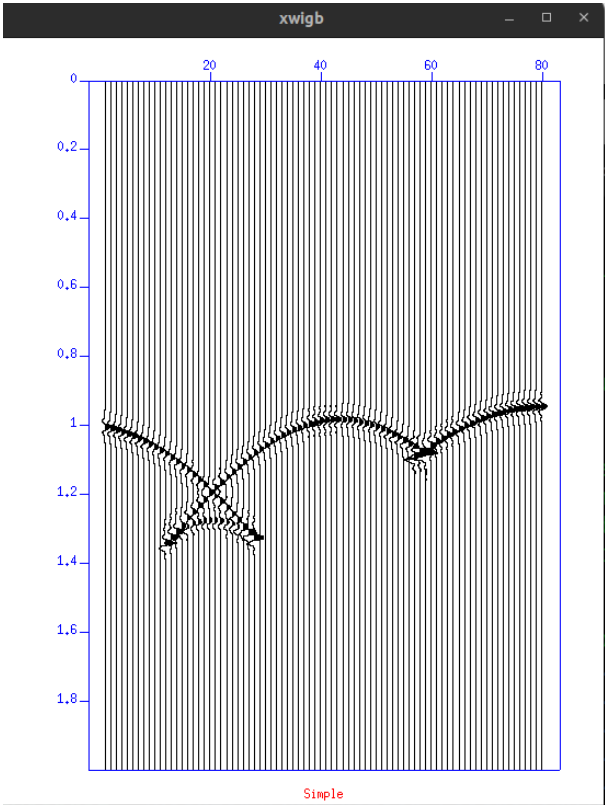
```
#!/bin/bash
surange<simple.su
suxwigb<simple.su xcur=3 title="Simple" &
sleep 2
sustolt<simple.su cdpmin=1 cdpmax=80 dxcdp=40 vmig=2000 tmig=0.0> stolt.simple.su
suxwigb<stolt.simple.su xcur=3 title='Stolt migration of simple data' &
sleep 2
suspikes nspk=4 nt=501 ntr=80 dt=.004 ix1=40 it1=63 ix2=40 it2=175 ix3=40 it3=40 it3=330 ix4=40 \
it4=420 | sushw key=cdp a=1 b=1 > spike_simple.su
suxwigb<spike_simple.su &
sustolt<spike_simple.su vmig=2000 tmig=0.0 dxcdp=40 cdpmin=1 cdpmax=80>stolt.spike.su
suxwigb<stolt.spike.su title="Stolt Migration after spike" &
sleep 2
#-----#
#-----Gazdag or Phase shift migration-----#
#-----#
sugazmig<simple.su dx=40 vmig=2000 tmig=0.0>gaz.simple.su
suxwigb<gaz.simple.su title="Gazdag Migration" &
#Migration by phase-shift migration
sumigps<simple.su dx=40 vmig=2000 tmig=0.0>ps.simple.su
suxwigb<ps.simple.su title="Phase Shift Migration" &
#Claerbout finite-difference migration
sumigfd< simple.su dx=40 dz=12 nz=150 vfile=vel.fdmig.simple >fd.simple.su
suxwigb<fd.simple.su title="Claerbout finite-difference migration" &
sleep 2
#Ristow and Ruhl's Fourier finite-difference migration
```

```

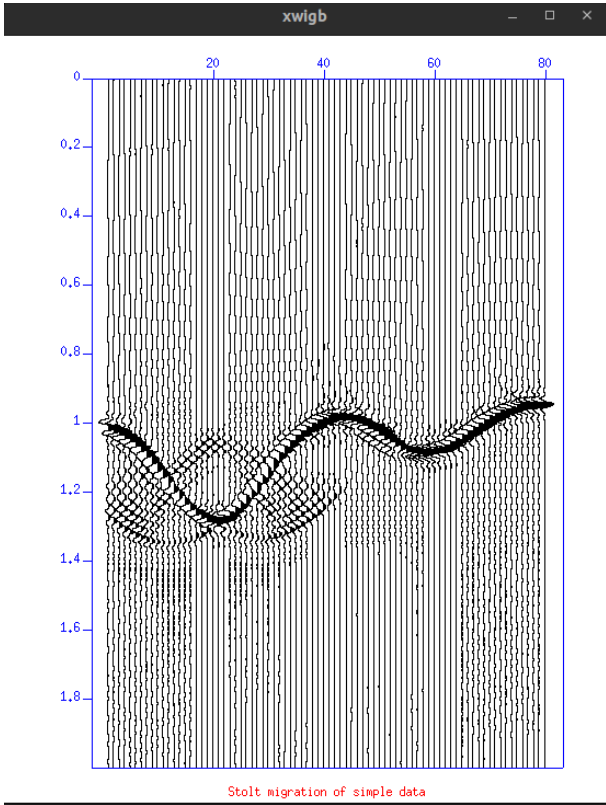
4 sumigffd< simple.su dx=40 dz=12 nz=150 vfile=vel.fdmig.simple >ffd.simple.su
5 suxwib<ffd.simple.su xcur=3 title="Ristow and Ruhl's Fourier finite-difference migration" &
6 sleep 2
7 #Stoffa's split-step migration
8 sumigss< simple.su dx=40 dz=12 nz=150 vfile=vel.fdmig.simple >ss.simple.su
9 suxwib<ss.simple.su title="Stoffa's split-step migration" &
10 sleep 2
11 #Gazdag Phase-shift Plus Interpolation migration
12 sumigpspi< simple.su dx=40 dz=12 nz=150 vfile=vel.fdmig.simple >pspi.simple.su
13 suxwib<pspi.simple.su title="Gazdag Phase-shift Plus Interpolation migration" &
14
15 #-----#
16 #-----Kirchhoff Migration of zero offset data-----#
17 #-----#
18 surange<simple.su
19
20 #Initially if the data do not have sx and gx then do the following to fix the sx & gx of simple.su dat
21 sushw < simple.su key=sx,gx a=0,0 b=40,40 > simple.su
22
23 #For Kirchhoff migration, we need two main scripts: Rayt2d.simple to generate travelttime tables
24 #and Kdmig2d.simple to perform Kirchhoff migration on the simple.su data.
25 #This will generate the travelttime table file tfile.simple and display a travelttime movie
26 ./Rayt2d.simple
27 #This performs Kirchhoff migration on simple.su and saves the output as kdmig.simple.su.
28 ./Kdmig2d.simple
29 xmovie < tfile.simple clip=3 n1=501 n2=80 loop=1 title="Traveltime Table" &
30 #View the Kirchhoff Migration Output
31 suximage < kdmig.simple.su title="Kirchhoff Migration Output" &
32
33 #working with real data(seismic3.su)
34 surange < seismic3.su
35 suximage < seismic3.su perc=99 verbose=1 $ title="Real data " &
36
37 #-----#
38 #-----Stolt Migration using Stoltmig Script-----#
39 #-----#
40 #Run Stoltmig to generate stolt.seis.su:
41 ./Stoltmig
42 #Convert Stolt-Migrated Data to Depth Section
43 #Run Suttoz.stolt to generate stolt.depth.seis.su:
44 ./Suttoz.stolt
45 #To visualize the original, migrated time, and depth sections
46 suximage < seismic3.su clip=.2 title="Stacked Data" &
47 suximage < stolt.seis.su clip=.2 title="Stolt Time Section" &
48 suximage < stolt.depth.seis.su clip=.2 title="Stolt Depth Section" &
49 #-----#
50 #-----Phase-Shift Migration using PSmig Script-----#
51 #-----#
52 #Run PSmig to generate ps.seis.su:
53 ./PSmig
54 #Convert Phase-Shift Migrated Data to Depth Section
55 #Run Suttoz.psmig to generate ps.depth.seis.su:
56 ./Suttoz.psmig
57 #View the Phase-Shift Migration Results
58 suximage < ps.seis.su clip=.2 title="PS Time Section" &
59 suximage < ps.depth.seis.su clip=.2 title="PS Depth Section" &

```

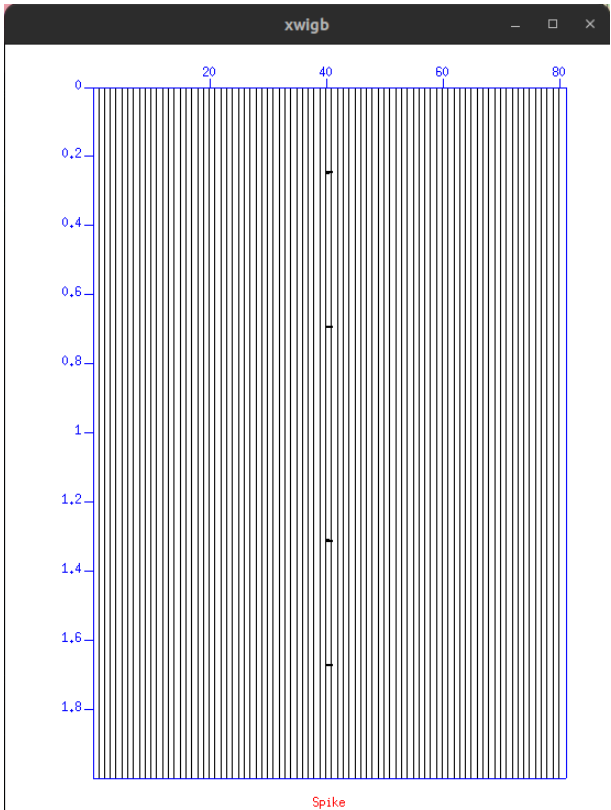
Results:



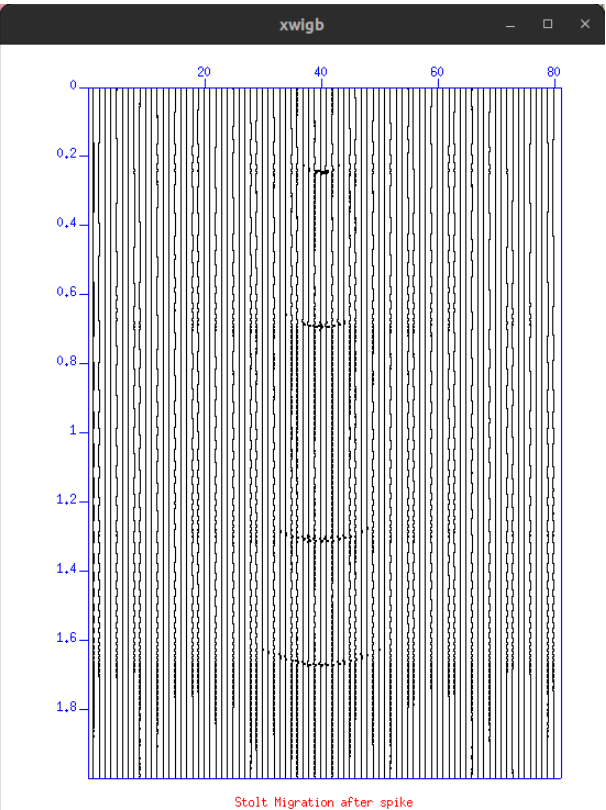
(a) Plot of **Simple.su**



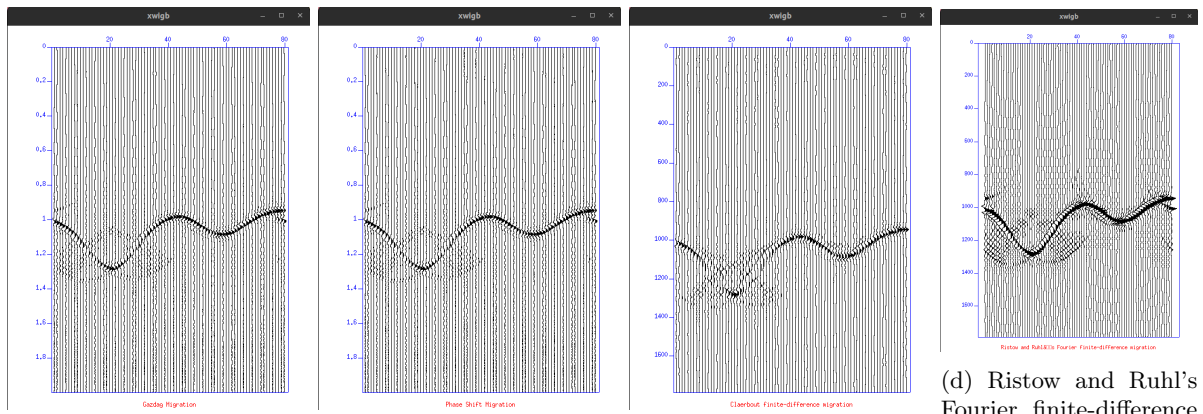
(b) Stolt migration of simple data



(c) Spike Plot.



(d) Stolt-migration-after-spike

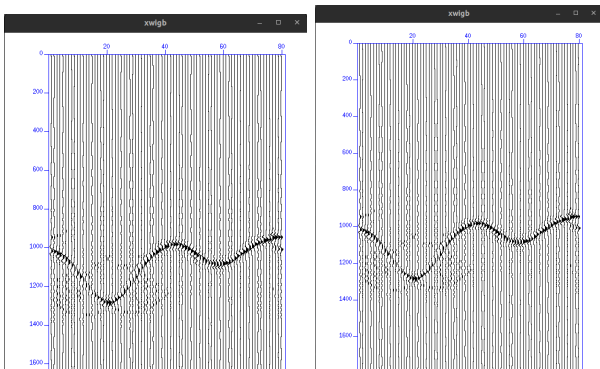


(a) Gadzag migration

(b) Phase shift migration

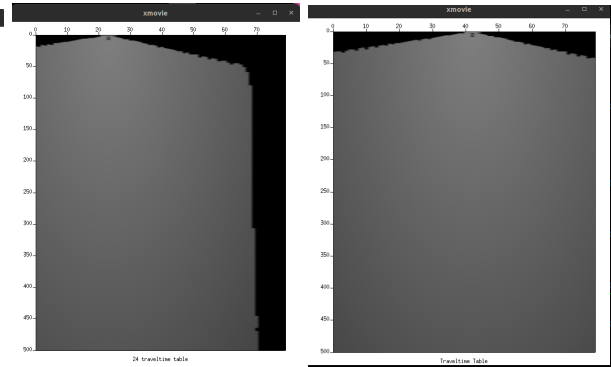
(c) Claerbout-finite-diff-mig

(d) Ristow and Ruhl's Fourier finite-difference migration.

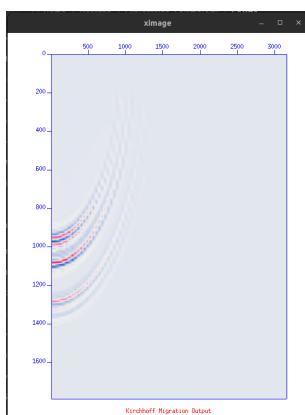


(e) Stoffa's-split-step-mig.

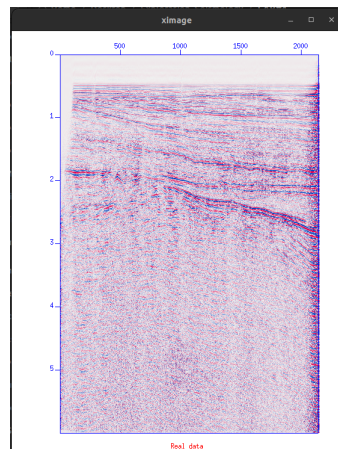
(f) Gazdag-ps-plus-interpolation-mig.



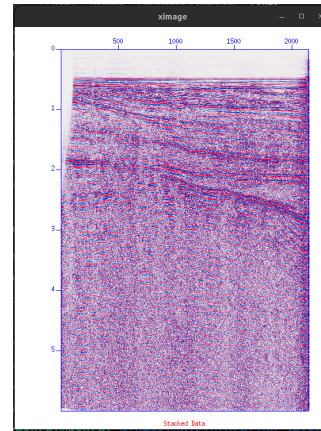
(g) travel-time table file (h) Kirchhoff migration on traveltime movie.



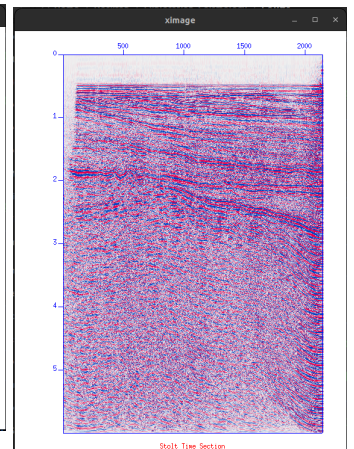
(i) Kirchhoffs migration output.



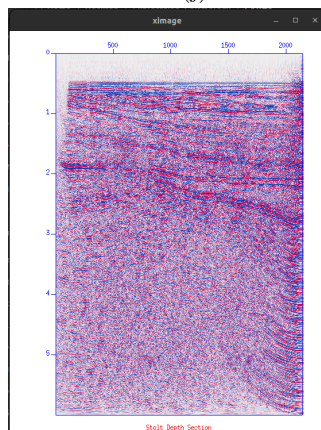
(j) Plot of seismic3 data.



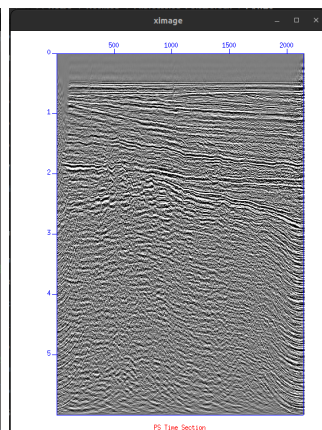
(k) Stacked data



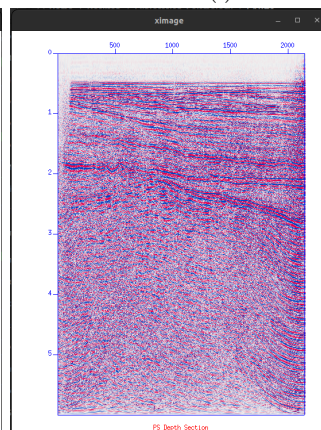
(l) stolt-time-section.png



(m) Stolt-depth-section.



(n) ps-time-section..



(o) PS Depth Section.

Figure 2: Results of various stages of Migration.